Rajiv Sambharya

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Academic Positions

- **Assistant Professor, Texas A&M University** Aug 2026–Present
  Industrial and Systems Engineering
- **Postdoctoral Researcher, University of Pennsylvania** Sep 2024–Jul 2026
  Electrical and Systems Engineering
  Supervision: George Pappas

Education

- **Ph.D., Princeton University** Sep 2019–Sep 2024
  Operations Research and Financial Engineering
  Thesis: Learning to Accelerate Optimization Algorithms with Guarantees
  Supervision: Bartolomeo Stellato
- **M.Sc. University of California - Berkeley** Sep 2017–May 2018
  Electrical Engineering and Computer Science
  Supervision: Laurent El Ghaoui
- **B.Sc. University of California - Berkeley** Sep 2013–May 2017
  Electrical Engineering and Computer Science

Research Interests

- **Data-driven computational tools for decision-making**
  Theory: optimization, machine learning, control theory.
  Methodology: machine learning for optimization, real-time optimization, optimization-based control, learning for control, statistical learning theory, computer-assisted optimization analysis.
  Applications: autonomous systems, signal processing, robotics, power systems, data science, operations research, finance.

Honors and Awards

- INFORMS Computing Society Prize, *INFORMS Computing Society* 2026
  Honorable mention. For research on learning to optimize with statistical performance guarantees.
- McGraw Teaching Fellow, *McGraw Center for Teaching and Learning, Princeton* 2022–2023
  Led the comprehensive 3-hour orientation for new teaching assistants.
- Excellence in Teaching Award, *Princeton School of Engineering and Applied Science* 2021
  Top award winner in engineering, for teaching ORF307 and ORF522.

Publications

Journal Articles

J1 R. Sambharya and B. Stellato, "Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization," *SIAM Journal on Mathematics of Data Science*, vol. 8, no. 3, pp. 649–676, 2026.
INFORMS Computing Society Prize, Honorable Mention

J2 R. Sambharya and B. Stellato, "Data-Driven Performance Guarantees for Classical and Learned Optimizers," *Journal of Machine Learning Research*, vol. 26, no. 171, pp. 1–49, 2025.
INFORMS Computing Society Prize, Honorable Mention

J3 R. Sambharya, G. Hall, B. Amos, and B. Stellato, "Learning to Warm-Start Fixed-Point Optimization Algorithms," *Journal of Machine Learning Research*, vol. 25, no. 166, pp. 1–46, 2024.
INFORMS Computing Society Prize, Honorable Mention

Conference Proceedings

C1 R. Sambharya, G. Hall, B. Amos, and B. Stellato, "End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization," in *Proceedings of The 5th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 211, PMLR, 2023, pp. 220–234.

Preprints

P1 R. Sambharya, S. C. Anand, and G. Pappas, "Verifying Performance, Stability, and Feasibility of Inexact Non-linear Model Predictive Controllers," 2026.

P2 R. Sambharya, N. Matni, and G. Pappas, "Verification of Sequential Convex Programming for Parametric Non-convex Optimization," 2025.

P3 R. Sambharya, J. Bok, N. Matni, and G. Pappas, "Learning Acceleration Algorithms for Fast Parametric Convex Optimization with Certified Robustness," 2025.

P4 A. Askari, G. Negiar, R. Sambharya, and L. El Ghaoui, "Lifted Neural Networks," 2018.

Thesis

T1 R. Sambharya, "Learning to Accelerate Optimization Algorithms with Guarantees," Ph.D. dissertation, Princeton University, 2024.

Working Papers

W1 S. Anand, S. Noorani, R. Sambharya, and G. Pappas, *Conformal Performance Guarantees for Learning to Optimize Mixed-Integer Programs.*

Talks

Verification of Sequential Convex Programming for Parametric Non-Convex Optimization

- INFORMS Optimization Society — Atlanta, Georgia, Mar 2026

Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization with Certified Robustness

- Engineering Technology & Industrial Distribution Seminar — Texas A&M University, Sep 2026
- Robotics Graduate Student Organization Seminar — University of Delaware, Sep 2026
- SIAM Conference on Optimization — Edinburgh, Scotland, Jun 2026
- Industrial and Systems Engineering Department Seminar — Texas A&M University, May 2026
- Industrial and Systems Engineering Department Seminar — North Carolina State University, Jan 2026
- INFORMS Annual Meeting — Atlanta, Georgia, Oct 2025
- International Conference on Continuous Optimization — Univ. of Southern California, Jul 2025

Data-Driven Performance Guarantees for Classical and Learned Optimizers

- NeurIPS — San Diego, California, Dec 2025
- International Symposium on Mathematical Programming — Montreal, Canada, Jul 2024
- Optimization Learning and Control Workshop (Poster) — Princeton University, Jun 2024

- INFORMS Optimization Society — Rice University, Mar 2024
- Conference on Information Sciences and Systems — Princeton University, Mar 2024

Learning to Accelerate Optimizers with Guarantees

- Thesis defense — Princeton University, Sep 2024
- **George Pappas's group** — University of Pennsylvania, May 2024
- **REALM lab** — MIT, Mar 2024
- **Computational Robotics Group** — Harvard University, Mar 2024

Learning to Warm-Start Fixed-Point Optimization Algorithms

- Yale Robotics Seminar — Yale University, Dec 2023
- INFORMS Annual Meeting — Phoenix, Arizona, Oct 2023
- Modeling and Optimization: Theory and Applications — Lehigh University, Aug 2023

End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization

- Learning for Dynamics and Control (Poster) — University of Pennsylvania, Jun 2023
- NYC Operations day (Poster) — Columbia University, May 2023
- INFORMS Annual Meeting — Indianapolis, Indiana, Oct 2022

Accelerating Non-Convex Optimization via Learned Sequential Convexifications

- International Conference on Continuous Optimization — Lehigh University, Jul 2022

Learning for Real-Time Semidefinite Optimization

- INFORMS Annual Meeting — Anaheim, California (hybrid), Oct 2021

Teaching

Texas A&M University (as Instructor)

- DAEN489/ISEN489: Special Topics in Computational Optimization (BSc level) — Spring 2027
  - **Topics:** linear algebra and calculus review, optimization basics, convex optimization, solving linear systems, equality-constrained least squares, linear programming, duality, Newton's method, interior-point methods, robust optimization, semidefinite programming, and integer programming.
- DAEN429: Data Analytics II (BSc level) — Fall 2026
  - **Topics:** linear algebra and multivariable calculus review, optimization, convex optimization, gradient descent, linear regression, linear classifiers, neural networks, generalization for neural networks, deep learning algorithms, convolutional neural networks, recurrent neural networks, attention and transformers, PCA and autoencoders, and variational autoencoders.

Princeton University (as Teaching Assistant)

- ORF499: Senior Thesis — Spring 2024
- ORF498: Senior Thesis — Fall 2023
- ORF363: Computing and Optimization for the Physical and Social Sciences — Spring 2023
- ORF387: Networks — Fall 2022
- ORF522: Linear and Nonlinear Optimization (PhD level) — Fall 2021
  Princeton Excellence in Teaching Award
- ORF307: Optimization — Spring 2021 (Head TA), 2022
  Princeton Excellence in Teaching Award
- ORF455: Energy and Commodities Markets — Fall 2020

McGraw Center for Teaching and Learning, Princeton (as Teaching Fellow)

- Led the comprehensive 3-hour orientation for new teaching assistants — 2022–2023

Mentoring

- **Research supervision**: Graduate students Tesshu Fujinami and Sima Noorani (Penn) 2025–Present
- **Research group leader**: Led weekly research group of 11 senior thesis students (Princeton) 2023–2024

Software

- Verification of Sequential Convex Programming for Parametric Non-convex Optimization
  github.com/rajivsambharya/verify_scp
- Learning Acceleration Algorithms for Fast Parametric Convex Optimization with Certified Robustness
  github.com/rajivsambharya/learn_algo_steps_robust
- Learning Algorithm Hyperparameters for Fast Parametric Convex Optimization
  github.com/stellatogrp/learning_algorithm_hyperparameters
- Data-Driven Performance Guarantees for Classical and Learned Optimizers
  github.com/stellatogrp/data_driven_optimizer_guarantees
- Learning to Warm-Start Fixed-Point Optimization Algorithms
  github.com/stellatogrp/l2ws
- End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization
  github.com/stellatogrp/l2ws_qp

Industry Experience

- Machine Learning Engineer at Linc Global Sunnyvale, California, Jul 2018–Jul 2019
- Software Engineering Intern at Amazon Seattle, Washington, Jun 2016–Aug 2016

Service

- Presentation Competition Judge, *INFORMS Texas A&M Student Chapter* 2026
  Judged undergraduate and graduate student talks at the ISEN Presentation Competition.
- Session Organizer, *INFORMS Optimization Society Conference* 2026
  Organized the session "Data-Driven Algorithm Design and Analysis for Parametric Optimization."
- Group Meeting Organizer, *Pappas Group, University of Pennsylvania* 2025–Present
  Organize the weekly research group meeting.
- Program Committee Member, *ScaleOpt Workshop at NeurIPS* 2025
  Reviewed workshop submissions.
- Session Organizer, *INFORMS Optimization Society Conference* 2024
  Organized the session "Machine Learning for Optimization."
- Seminar Organizer, *Princeton Optimization Seminar, Princeton University* 2022–2023
  Organized the optimization seminar series.

Peer Review

- IEEE Transactions on Automatic Control (x1)
- NeurIPS (x2)
- IEEE Conference on Decision and Control (x1)
- Conference on Learning Theory (x2)
- Learning for Dynamics and Control (x5)
- Integer Programming and Combinatorial Optimization (x1)

Technical Skills

- **Programming languages**: Python, Matlab, Julia, R, C, Java, SQL, HTML
- **Tools**: Git, LaTeX, Slurm, GPU, JAX, PyTorch, Tensorflow